



Review

Influence of cooking techniques on food quality, digestibility, and health risks regarding lipid oxidation

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ABSTRACT

Foods undergo various physical and chemical reactions during cooking. Boiling, steaming, baking, smoking and frying are common traditional cooking techniques. At present, new cooking technologies including ultrasonic-assisted cooking, vacuum low-temperature cooking, vacuum frying, microwave heating, infrared heating, ohmic heating and air frying are widely studied and used. In cooking, lipid oxidation is the main reason for the change in lipid quality. Oxidative decomposition, triglyceride monomer oxidation, hydrolysis, isomerization, cyclization reaction and polymerization occurred in lipid oxidation affect lipids' quality, flavor, digestibility and safety. Meanwhile, lipid oxidation in cooking might cause the decline of lipid digestibility and increase of health risks. Compared with the traditional cooking technology, the new cooking technology that is milder, more uniform and faster can reduce the loss of lipid nutrition and produce a better flavor. In the future, the combination of various cooking technologies is an effective strategy for families to obtain healthier food.

1. Introduction

Cooking is an important part of the home's food preparation. A series of physical and chemical changes occur during cooking, which ultimately change the appearance, sensory quality, nutritional value and safety of food. Traditional cooking techniques include frying, boiling, steaming and roasting. Nowadays, vacuum cooking, infrared heating, microwave heating and ultrasonic-assisted cooking technologies are rapidly developing. Different cooking techniques could affect the food properties by changing food surface temperatures, heat conduction dynamics. For instance, vacuum cooking preserves more chlorophyll, carotenoids, and phenols in vegetables than boiling, and exhibits less loss of antioxidant properties (Guillen, Mir-Bel, Oria, & Salvador, 2017). In addition to affecting the retention rate of nutrients, appropriate cooking conditions can improve the texture and taste of food. Previous study shows that the chicken breast obtained by cooking for 4 min in the oven at 130–150 °C presents the best tenderness and the lowest cooking loss (Barbanti & Pasquini, 2005). However, the inappropriate selection of cooking parameters in cooking may produce acrylamide, heterocyclic

amines, volatile aldehydes and other potential health hazards (Zhang, Saleh, Chen, & Shen, 2012). In a word, cooking technology leads to changes in the composition and structure of food, which have a comprehensive impact on food properties. Therefore, the selection of appropriate cooking techniques is extremely important.

Lipid is an important component of food, which provides necessary energy and nutrients for human body, and plays an important role in food sensory and flavor (Genovese, Caporaso, & Sacchi, 2021; Martins, Vicente, Pastrana, & Cerqueira, 2020). Nowadays, edible oils or foods rich in unsaturated fatty acids (UFAs) and essential fatty acids are recognized as lipids with high nutritional value, such as olive oil, camellia oil and deep-sea fish products (Flori et al., 2019; Khalili Tilami & Sampels, 2017; Li et al., 2022). However, the rich content of UFAs in edible lipids also leads to their fragile oxidative stability. The oil oxidation is main reason for oil deterioration. The spontaneous oxidation of UFAs and the oxidation reaction not only reduce lipids' nutritional value, but also produce unpleasant flavor and substances such as malondialdehyde, 4-hydroxy-2-nonenal and hexanal (Huang & Ahn, 2019). At present, the influence of cooking technology on lipids has

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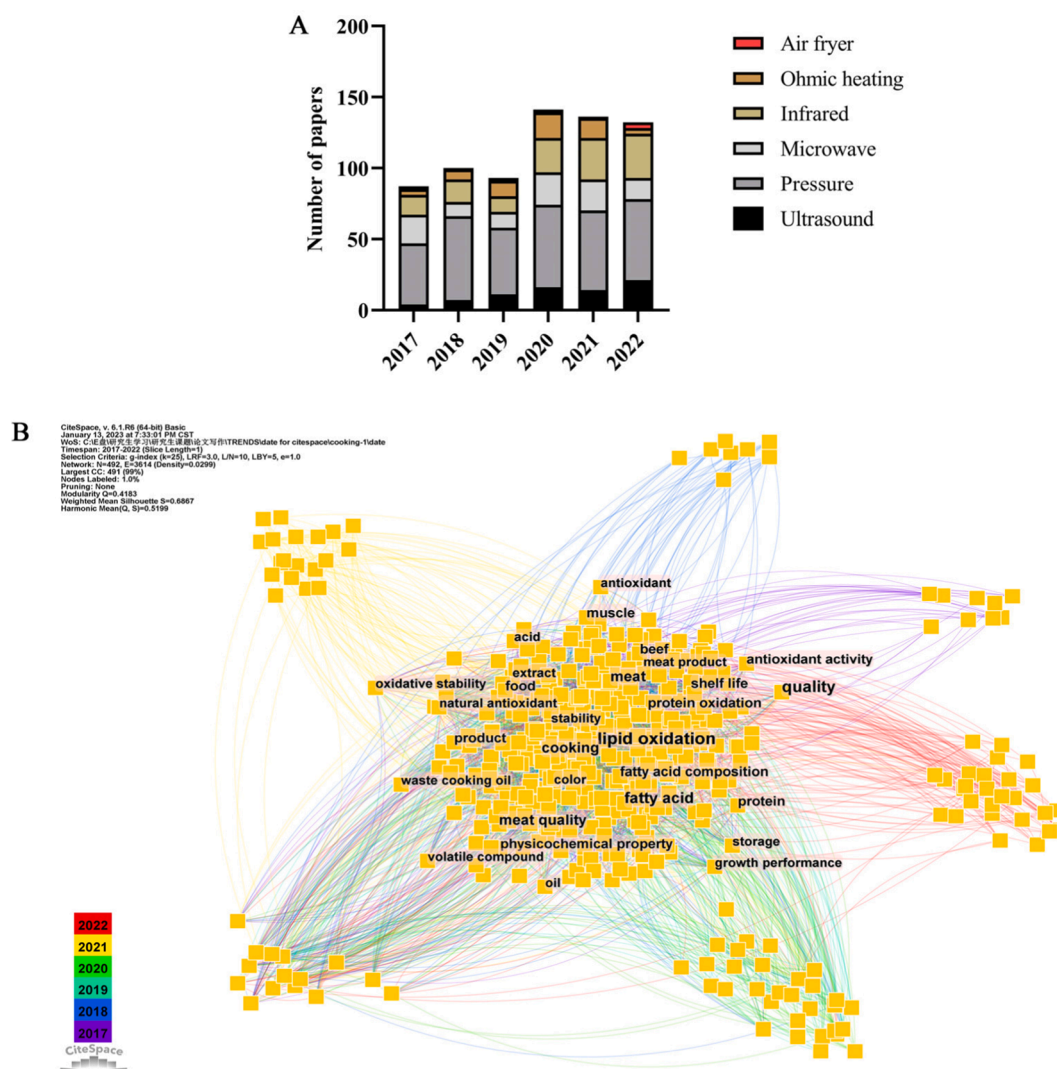


Fig. 1. CiteSpace software analysis (version: 6.1.R6) of the literature on the theme of “cooking” and “lipids” (2017–2022). (A) The relationship between the literature with the theme of “Emerging cooking technology” and the time of publication (2017–2022, GraphPad Prism 8.3.0), according to the analysis results of CiteSpace software. (B) CiteSpace software’s analysis of the literature keywords with the theme of “cooking” and “lipids”. The different color lines indicate different publication years, and the larger the font meant the higher the frequency of occurrence.

gradually become a research hotspot (Fig. 1). Among them, lipid oxidation, meat quality, flavor substances, food color, oxidation stability, shelf life and physicochemical properties have been continuously concerned (Fig. 1). In the past two years, studies on lipid replacement, pulsed electric field and anaerobic digestion have also increased. In the innovation of cooking technology, air frying, ohmic heating, infrared heating, microwave heating, pressure cooking and ultrasonic-assisted cooking also provide a new perspective for the study of lipid changes in cooking. From 2017 to 2022, researchers have been interested in exploring the impact of pressure on lipid cooking, and the use of ultrasonic-assisted and infrared heating technology has also become more frequent (Fig. 1). In all, it is of great significance to discuss the effects of different cooking techniques on lipids to find out the changes in nutritional value, safety and flavor of lipids in practical applications.

In this review, different cooking techniques and their effects on lipid oxidation, flavor, digestibility and safety were systematically discussed. This paper aimed to reveal the important influence of cooking technology on the quality and health characteristics of oils and fats, and provide a new perspective for the development and research of edible lipids in the future.

2. Cooking techniques

2.1. Traditional cooking techniques

According to the contact type of food raw materials and water, water-mediated cooking techniques can be divided into stewing and steaming. Stewing is a popular cooking method in Southeast Asia. In stewing, the ingredients are immersed in water and achieve protein denaturation and other cooking effects through water-mediated heat conduction. Compared to stewing, the food in steaming has no direct contact with water and enables low temperature cooking ($\leq 100^\circ\text{C}$) (Table 1). In steaming, the water is placed at the bottom of the pot and the food is placed in an overhead container that allows steam to pass through. When the pot is covered, heat transfer from the hot and humid air to the food is achieved. Roasting and smoking are traditional air-mediated cooking methods and widely applied in food processing. In roasting, food is processed at a temperature of $150\text{--}300^\circ\text{C}$ or even higher, accompanied by a series of physical and chemical changes, which in turn form unique flavor, color and aroma (Table 1) (Sruthi, Premjit, Pandiselvam, Kothakota, & Ramesh, 2021). Generally speaking, in the initial stage of roasting (about 160°C), the moisture on the surface of the food is rapidly evaporated and the migration of food moisture

Table 1
Examples of cooking technical parameters.

Cooking technology	Food matrix	Temperature	Time	Pressure (Pa)/Power (W)/Frequency (kHz)/Wavelength (um)/Relative humidity/Electric field strengths	References
Stewing	Beef	100 °C	2 h, 3 h, 4 h, 5 h, 6 h	/	(Wang et al., 2020)
Steaming	Fish (<i>Micropterus salmoides</i>)	100 °C and 75 °C	100 °C for 12 min and then 75 °C for 12 min; 75 °C for 24 min and then 100 °C for 8.9 min	/	(Wang, Bao, Wang, Chen, & Zhou, 2020)
Roasting	Round steak	200 °C	10 min on both sides	/	(Al-Dalali, Li, & Xu, 2022)
Smoking	Nile tilapia (<i>Oreochromis niloticus</i>)	4 °C	18 h	Relative humidity (90 %)	(Ruiz-Alonso, Girón-Hernández, López-Vargas, Muñoz-Ramírez, & Simal-Gandara, 2021)
Deep-fat frying	Skin of tilapia (<i>Oreochromis</i> spp.)	180 °C	2, 4, 6 and 8 min	/	(Fang, Huang, & Sung, 2021)
Electrostatic frying	Skin of tilapia (<i>Oreochromis</i> spp.)	180 °C	2, 4, 6 and 8 min	/	(Fang et al., 2021)
Ultrasound-assisted stewing	Beef	100 °C	2 h	0, 63.00, 74.76, or 93.33 W dm ⁻³ , 20 kHz	(Zhang, Zhang, Zou, & Zhang, 2021)
Ultrasound-assisted frying	Beef meatballs	160 °C	12 min	0, 200, 400, 600 and 800 W, 20 kHz	(Zhang et al., 2020)
Ultrasound and microwave-assisted vacuum frying	Mushroom slice	90 °C	2, 4, 6, 8, 10, 12 and 14 min	0.015 MPa, 1000 W (microwave), 300 or 600 W (ultrasound)	(Devi et al., 2021)
Microwave heating	Rabbit meat	/	5 min	1000 W	(Abdel-Naeem, Sallam, & Zaki, 2021)
Microwave-assisted vacuum frying	Mushroom slice	90 °C	2, 4, 6, 8, 10, 12 and 14 min	0.015 MPa, 1000 W	(Devi et al., 2021)
Air frying	Skin of tilapia (<i>Oreochromis</i> spp.)	180 °C	6, 8, 10, 12 and 14 min	/	(Fang et al., 2021)
Sous-vide cook	Beef steak	55 °C	45 or 120 min	vacuum	(Sun, Rasmussen, Cavender, & Sullivan, 2019)
Vacuum frying	Skin of tilapia (<i>Oreochromis</i> spp.)	120 °C	4, 8, 12, 16, 20 and 24 min	0.058Mpa	(Fang et al., 2021)
Infrared heating	Lamb patties	240 °C	17, 18, 19, 20 and 21 min	/	(Suleman, Hui, Wang, Liu, & Zhang, 2020)
Infrared frying	Apple slices	160 °C	2, 4, 6, 8, 10 and 12 min	wavelength: 2.0–2.5 μm; 2000, 2500 and 3000 W	(Su et al., 2022)
Ohmic cooking	Instant rice cake soup	100 °C	60, 80, 100 and 120 s	electric field strengths (9, 12, 15, 18 V/cm)	(Soisungwan, Khampakool, You, & Park, 2020)

from the center to the surface may occur at the same time. Further, chemical changes such as hydrolysis, oxidation, thermal decomposition and condensation of proteins, lipids and starches in food occurred when the temperature reached 190 °C (Sruthi et al., 2021). Moreover, smoking is another air-mediated cooking technology, which make the food smoked with heat or aroma substances (Table 1). Interestingly, cold smoking is often carried out at 12–25 °C to enhance the flavor of food, especially food of animal origin (Unlusauyin, Kaleli, Bilgin, Izci, & Gunlu, 2006), while hot smoking usually maintains the temperature at 40–100 °C to achieve the purpose of improving sensory quality and cooking (Unlusauyin et al., 2006).

Plant oils and animal fats are commonly used as media for cooking food due to their large specific heat capacity and high boiling point (Vitrac, Trystram, & Raoult-Wack, 2000). Frying, a cooking technique in which food is dipped or floated on the surface of oil and generally controlled temperature at 150–190 °C, is popular for its convenience and palatability (Table 1) (Hosseini, Ghorbani, Meshginfar, & Mahoonak, 2016; Zhang et al., 2012). In the process of frying, heat and material transfers occur simultaneously between the food and the oil: the temperature of the food rises rapidly, the moisture in food is rapidly evaporated and replaced by the oil (Vitrac et al., 2000). Meanwhile, the surface or interior of food will expand, crack and shrink (Vitrac et al., 2000). However, heterocyclic amines, acrolein, free fatty acids, cyclic fatty acid monomers, trans fatty acids, and triacylglyceride polymers produced during frying not only affect the sensory quality of fried foods but also may lead to health risks (Zhang et al., 2012). Besides, most of the lipids in fried foods come from fried oil, which must be carefully considered in the study of lipid fate during cooking. The type of food, oil composition, frying time, temperature, oxygen content and utensils will

affect the above side reactions (Bordin, Kunitake, Aracava, & Favaro-Trindade, 2013). Therefore, the future focus area may include find healthy frying assist technology, optimal frying condition.

2.2. New cooking techniques

2.2.1. Ultrasonic-assisted cooking technology

Ultrasonic refers to sound waves with a frequency greater than 20 kHz, which are widely used for extraction, dehydration, enhancement of physicochemical properties, and microbial inactivation in food industry (Jadhav & Annature, 2021). Briefly, the microbubble effect is generated as ultrasonic waves are conducted in the liquid medium, and the continuous action of ultrasonic waves could cause microbubbles collapse to form turbulent flow, resulting in instantaneous mechanical shear force, high temperature and high pressure. Therefore, ultrasonic treatment can improve the mass transfer efficiency of the system (Jadhav & Annature, 2021). At present, ultrasound has played an active role in frying and boiling (Antunes-Rohling et al., 2018; Zou et al., 2018).

2.2.2. Pressure-regulated cooking techniques

Pressure is an important condition that affects mass transfer and boiling point. Therefore, adjusting the pressure to alter the cooking process is a viable strategy (Table 1). Sous-vide is the technique of cooking food inside a vacuum bag at 65–95 °C for 1–7 h (Renna, Giannella, Giannino, & Santamaria, 2014). Lower temperature and oxygen content reduce the loss of nutrients, moisture and color, inhibit the oxidation of lipids and proteins, and further hindered the production of harmful substances such as polycyclic aromatic hydrocarbons and heterocyclic amines (Kathuria, Dhiman, & Attri, 2022). Vacuum frying

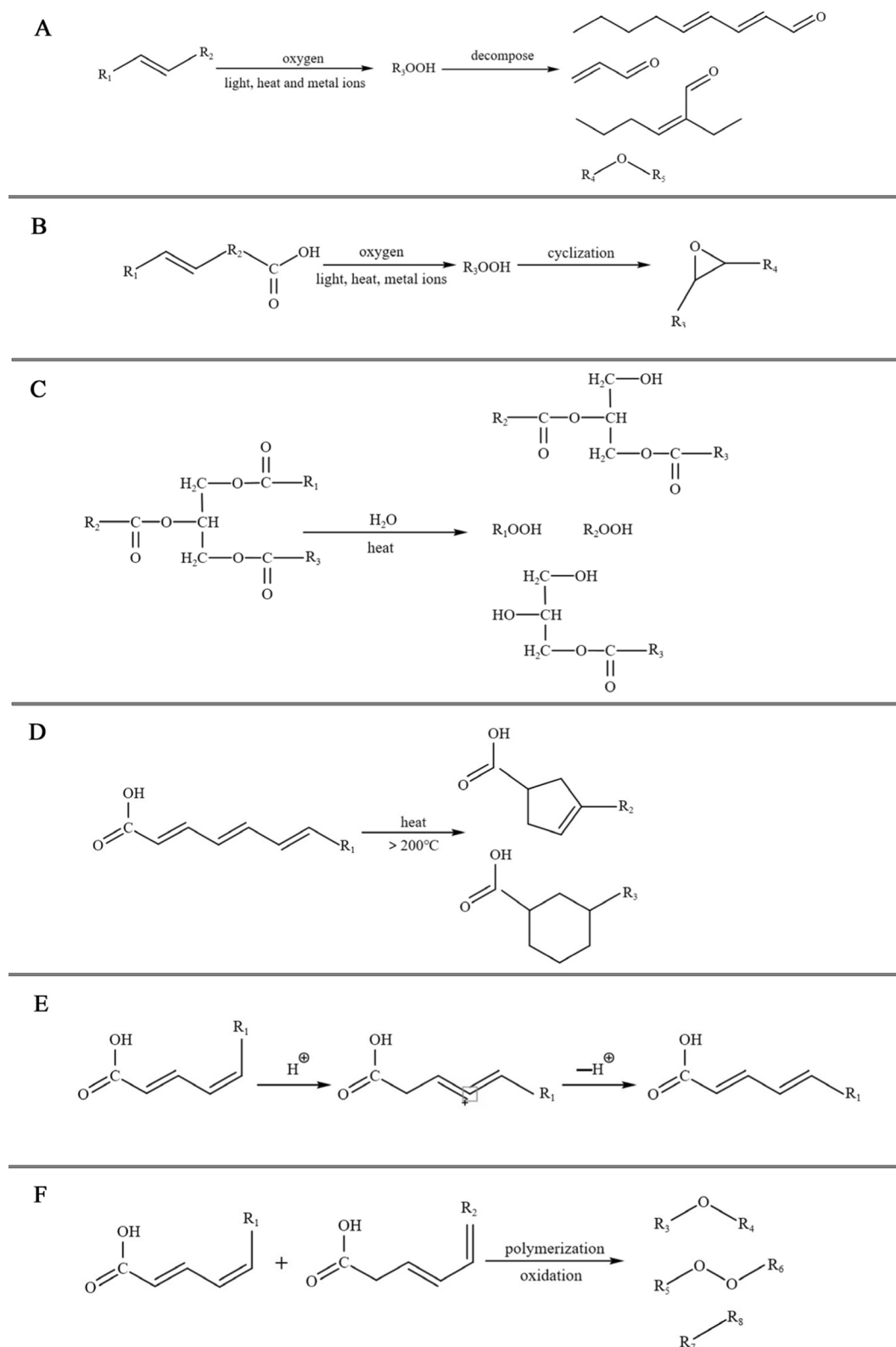


Fig. 2. Oxidation of lipids in cooking. (A) Oxidative decomposition. (B) Fatty acid monomer oxidation. (C) Hydrolysis. (D) Isomerization. (E) Fatty acid monomer cyclization reaction. (F) Thermal/oxidative polymerization.

refers to the frying technique performed at a pressure below 6.65 kPa (Andres-Bello, Garcia-Segovia, & Martinez-Monzo, 2011). Compared with traditional frying, the boiling point of water in the food is lower in vacuum condition, which realizes the rapid evaporation of water at a lower temperature and achieves the purpose of frying. Thus, vacuum

frying greatly reduces nutrient loss, oil absorption, production of harmful substances (Belkova et al., 2018; Devi, Zhang, Ju, & Bhandari, 2020). Conversely, increasing cooking pressure is also an effective technique for improving food quality. Pressure frying is carried out at a pressure above atmospheric pressure. Under high pressure, the

evaporation of water inside the food is inhibited, thereby improving the water content and tenderness of the fried food, especially suitable for frying animal muscles (Su et al., 2022).

2.2.3. Microwave cooking technology

Microwaves are non-ionizing radiation with frequencies between 300 MHz and 300 GHz. In microwave heating, the dielectric properties of food and the penetration depth of microwaves are important factors affecting the heating efficiency (Table 1) (Chandrasekaran, Ram-anathan, & Basak, 2013). A previous study recorded microwave technology combined with traditional frying cooking can achieve higher heat and mass transfer efficiencies (Nasiri, Mohebbi, Yazdi, Hossein, & Khodaparast, 2011).

2.2.4. Infrared cooking technology

Infrared is electromagnetic waves with wavelengths between 0.5 and 1000 μm , which are divided into near infrared (0.78–1.4 μm), mid infrared (1.4–3 μm), and far infrared (3–1000 μm) according to wavelength. Compared with traditional cooking technologies, infrared heating is uniform, rapid, and energy-saving while causing only limited food oxidation, which is widely used in the cooking (Table 1) (Ogundele & Kayitesi, 2019; Sruthi et al., 2021).

2.2.5. Ohmic heating cooking

Ohmic heating is a new heating technology that uses an alternating electric field to generate heat. Specifically, the water and salt contained in the food become electrical conductors. When the food is placed in an alternating electric field, the ions in the food repeatedly move to the electrodes with opposite charges, resulting in collision and heat generation (Alkanan, Altemimi, Al-Hilphy, Watson, & Pratap-Singh, 2021). At present, ohmic heating is widely used in fruit juice cooking and has only limited adverse effects on the nutrition, function and sensory characteristics of food (Table 1) (Alkanan et al., 2021). However, the application of ohmic heating in other food needs more effort.

2.2.6. Air fryer cooking techniques

Air fryer technology is an emerging alternative to traditional frying technique. Air fryer technology uses hot air heat conduction instead of traditional oil heat conduction (Table 1). This technology can make a small amount of oil evenly dispersed in the hot air and then contact with the surface of the food. The atomization of the oil droplets improves the heat transfer efficiency and uniformity, and reduces the quality loss of the food during the frying process (Liu et al., 2022).

2.2.7. Composite new cooking technology

Recently, the combined usages of various new cooking technologies show synergy in improving food quality and cooking efficiency. When ultrasonic treatment and vacuum frying are combined, because ultrasonic speeds up the mass transfer process, the frying temperature, oil absorption rate and frying time are reduced, and the dehydration rate, nutritional value and sensory properties of food are improved (Sosa-Morales, Solares-Alvarado, Aguilera-Bocanegra, Munoz-Roa, & Cardoso-Ugarte, 2022). Besides, vacuum frying is combined with microwave processing, which significantly accelerated the evaporation of water in frying, and the treated samples had more uniform void distribution, better appearance and lower oil content (Shi, Zhang, & Yang, 2019). In short, the combined use of various cooking technologies plays a beneficial role in improving cooking efficiency, improving sensory characteristics and nutritional value (Islam, Zhang, & Fan, 2019; Su, Zhang, Chitrakar, & Zhang, 2020).

3. Changes in the composition of lipids in cooking

The composition and distribution of fatty acids in food lipids affect their cooking stability. The oxidation, hydrolysis, polymerization and other types of reactions might undergo in a heated environment,

eventually changing the composition of lipids and affecting the quality of food (Fig. 2).

3.1. Oxidative decomposition products

Under high temperature conditions, auto-oxidation and thermal oxidation of lipids are the main causes of oil rancidity. The oxidative decomposition of oil is a free radical reaction, which is classically divided into free radical initiation, propagation and termination. In the presence of oxygen, light, heat and metal ions can induce the generation of acyl radicals and further formation of peroxy radicals. The free radical attacks the glycerol molecule to snatch hydrogen to form hydroperoxide. Hydroperoxide is extremely unstable and easily causes β -scission homolysis of C—C bond, C—O bond and O—O bond in the glyceride molecule, and thus forms an alcohol with a smaller molecular weight, such as acrolein, hexenal, nonadienal and decenaldehyde (Fig. 2) (Zhang et al., 2012). These polar volatile substances contribute to the unpleasant flavor of heated lipids and may also cause health risks (Wu, Yen, & Sheu, 2001; Yildizbayrak, Orta-Yilmaz, Aydin, & Erkan, 2020).

3.2. Triglyceride monomer oxidation products

Under the attack of hydroperoxide, the C=C bond of unsaturated fatty acid triglyceride is easily opened and the group of ketone, epoxy, hydroxyl and aldehyde are introduced, leading to triglyceride monomer oxidation product formed (Fig. 2). Among them, triglycerides produced by C=C epoxidation are common products (Li et al., 2022). Chen et al. (2020) showed that polyunsaturated fatty acids (PUFAs)-rich flaxseed oil produced more triglyceride monomer oxides than peanut, canola, and soybean oils under the same cooking conditions.

3.3. Hydrolysate

The hydrolysis of oil and water under heating conditions is easy to generate glycerol, free fatty acids, monoglycerides and diglycerides (Gillatt, 2001). In the non-enzymatic hydrolysis reaction, the order of ester bond breaking may be affected by water content, temperature, the position and number of C=C and the length of the carbon chain of fatty acids (Zhang et al., 2012). In cooking, the rise of temperature may accelerate the hydrolysis of triglycerides, while the evaporation of water caused by temperature rise may also inhibit the hydrolysis of oil.

3.4. Isomerization product

Isomerization reaction is a primary path to produce toxic and harmful substances in cooking. Cyclic isomers and trans fatty acids are the main products of lipid isomerization. Among them, trans fatty acids are widely concerned because of their universality. Mechanistically, the rearrangement and migration of electrons in UFAs may lead to the cleavage and displacement of C=C bonds. Meanwhile, oxidative radicals attack the C=C bonds to form free radical adducts, which finally form the lower energy trans configuration through the β -elimination reaction (Fig. 2) (Zhang et al., 2012). The increase of cooking temperature and the extension of cooking time may promote the accumulation of isomerization products (Afaneh, Abbadi, Al-Rimawi, Al-Dabbas, & Sawalha, 2017).

3.5. Fatty acid monomer cyclization reaction product

Under a temperature of over 200 °C, UFAs are prone to the destruction of the C=C double bond to form intramolecular or inter-molecular cyclization reaction products, namely cyclic fatty acid monomers (Fig. 2) (Sebedio, Prevost, & Grandgirard, 1987). In terms of the formation mechanism, it's an intramolecular rearrangement reaction catalyzed by free radicals. According to the principle of double bond distribution in the molecule and the principle of minimum energy, cyclic

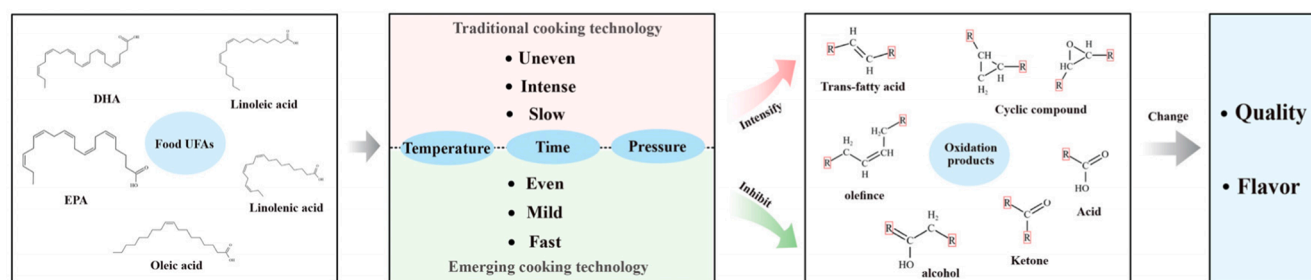


Fig. 3. Different cooking techniques lead to different levels of lipid oxidation and thus affect the quality and flavor of food.

fatty acid monomers usually present a five- or six-membered ring monocyclic and bicyclic structures (Zhang et al., 2012).

3.6. Thermal/oxidative polymerization products

Polymerization is a common reaction in oil heating, which deeply affects the color, texture and usability of oil (Fig. 2). In an aerobic high-temperature environment, two, three or more fatty acid glycerides are connected by C—C, C—O—C and C—O—O—C bonds to form non-polar polymers: dimers, trimers and oligomers of fatty acid glycerides (Zhang et al., 2012).

3.7. Reaction of lipids with other food components

In the real cooking process, starch, protein, minerals, phenols, cellulose and other components in food may react with lipids to influence the nutritional value of food quality. At present, many reviews discussed the formation mechanism and research progress of the interaction between lipid oxidation and Maillard reaction. In these reactions, food components interact to produce nitrogen and sulfur containing heterocyclic compounds, acrylamide, binary polymers, ternary polymers and quaternary polymers (Krishnan, Mondal, Thomas, Singh, & Praveen, 2021; Wang et al., 2020). Therefore, this paper focuses on the changes of food lipids during cooking, without considering the effects of other food components.

4. Effects of cooking techniques on lipid oxidation and flavor

Lipid oxidation is an important reaction of lipids during cooking and is affected by cooking conditions and lipid properties. Oxidative products such as aldehydes, ketones, alcohols, acids, and short-chain hydrocarbons profoundly affect food flavor and lipid availability, which in turn are related to food quality. Therefore, the effect of cooking techniques on lipid oxidation and flavor is a topic of concern (Fig. 3). In the diet, meat is an important source of saturated and unsaturated fatty acids, such as stearic acid, oleic acid, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) (Ekmekcioglu et al., 2018; Valsta, Tapanainen, & Mannisto, 2005). Meat is the main source of palmitic acid, stearic acid and oleic acid in the diet (Ekmekcioglu et al., 2018; Valsta et al., 2005). Moreover, edible oil is an important lipid source in the diet (Pitts, Dorling, & Pattie, 2007). Hence, this review mainly discusses the effects of cooking techniques on lipid oxidation and flavor of four kinds of foods rich in oil and fat (red meat, white meat, fish and edible oil, Table 2). According to industry habits, beef, pork and mutton are called red meat, while meat from poultry and turkey is called white meat (Keeton & Dikeman, 2017).

4.1. Effect of different cooking techniques on lipid in red meat

Red meat commonly refers to red beef, mutton, veal and pork, and is an important dietary component in the world (Linseisen et al., 2002). Lipid is an important component of red meat, which not only provides

essential nutrients but also impacts flavor. For example, 2-nonenal and 2,4-decenal are the main sources of meat flavor. They can be formed by the automatic oxidation of linoleic acid and arachidonic acid through hydroperoxide (Arshad et al., 2018). In addition, 1-octene-3-one, 2,4-nonadienal and 2-octenal produced by oil oxidation are contributors to cooking flavor (Arshad et al., 2018). Utama et al. (2018) compared the effects of boiling and oven-baking on lipid oxidation and aroma volatiles of beef. They found that the levels of total SFAs and malondialdehyde (lipid oxidation products) of beef were increased with the core temperature rising from 70 °C to 77 °C, while the content of total mono-unsaturated fatty acids (MUFAs) was decreased (Utama et al., 2018). Additionally, at the same temperature, cooking in the oven gave more volatile aldehydes to beef than boiling in water, which enhanced the aroma of beef (Utama et al., 2018). In another study, the loss of water in roast rib-eye steak might be the reason for the increase of SFAs, MUFAs and n-6 PUFAs content, while the absorption of 12:0 and 18:2n-6 in margarine used for frying by rib eye steak led to the increase of corresponding fatty acid content of steak (Gruffat, Bauchart, Thomas, Parafita, & Durand, 2021). Besides, the smoking time was proven to have an important impact on lipid oxidation and flavor of cold smoked bacon (Li, Nie, Liu, & Xu, 2021). Specifically, the cured meat smoked for 12 and 15 days presented a mature flavor, mainly containing volatile compounds such as aldehydes, ketones, phenols, esters and heterocyclic compounds. Its comprehensive content is high on the 12 days of smoking, which is different from the samples smoked for 15 days (Li et al., 2021).

In recent years, the effects of new cooking techniques on lipid oxidation and flavor have also aroused researchers' interest. Zou et al. (2018) evaluated the effect of ultrasound-assisted stewing (frequency 20 kHz) on the quality of spiced beef. They recorded that ultrasound treatment obviously increased the level of lipid oxidation of beef, which was independent of ultrasound power. Meanwhile, ultrasonic treatment enhanced the level of lipid oxidation, which might be accompanied by the increase in the variety and content of volatile flavor substances such as aldehydes, alcohols and ketones (Zou et al., 2018). Similarly, the effect of ultrasonic power (0, 200, 400, 600 and 800 W) on the quality of meatballs in ultrasonic-assisted frying (12 min, 160 °C) was studied (Zhang, Zhang, Wang, Xing, & Zhang, 2020). The results showed that ultrasonic treatment significantly increased the content of thiobarbituric acid reactants in meatballs, accelerated the oxidation of free fatty acids and the formation of volatile lipid oxidation products (Zhang et al., 2020). These changes affect the flavor of meatballs and are detected by electronic nose (Zhang et al., 2020). In addition to ultrasonic treatment, using high-pressure technology to pickle pork chops could increase the absorption of pork chops to pickle and lead to an increase in oleic acid level (O'Neill, Cruz-Romero, Duffy, & Kerry, 2019). The cured pork chops after steam cooking had a lower n-6:n-3 PUFAs ratio than the frying group, which may lead to better sensory quality of the steam cooked pork chops (O'Neill et al., 2019). Besides, the effects of vacuum low temperature cooking on the quality and flavor of braised pork were also evaluated. Compared with traditional cooking, the fat content and malondialdehyde value of vacuum low-temperature cooking samples were lower, but there was no significant difference in sensory flavor

Table 2

Effects of different cooking techniques on food quality: starting from lipid (2017–2022).

Food category		Cooking Technology	Influence of quality	References
Red meat	Hanwoo brisket	Oven-roasted (180 °C) or cooked in boiling water to final core temperature of 70 °C (medium) or 77 °C (well-done)	Boiling reduced more fat. The increase of cooking core temperature led to more loss of MUFAs and production of malondialdehyde.	(Utama et al., 2018)
	Buffalo Meat	Grilling, roasting, boiling, steaming and frying	Compared with steaming, barbecue showed higher malondialdehyde level.	(Klinhom, Klinhom, & Methawiwat, 2017)
	Spiced Beef	Ultrasonic assisted boiling (ultrasonic power is 0 W, 400 W, 600 W, 800 W and 1000 W (20 kHz, 120 min))	The levels of lipid oxidation and the contents of flavor substances such as aldehydes, alcohols and ketones were significantly increased by ultrasound.	(Zou et al., 2018)
	Meatballs (yellow cattle)	Ultrasonic-assisted frying (12 min, 160 °C) with ultrasonic power of 0, 200, 400, 600 and 800 W respectively	Ultrasonic treatment significantly increased the oil oxidation rate, reduced the content of free fatty acids and promoted the production of volatile substances.	(Zhang et al., 2020)
	Beef	Frying, grilling, braising or roasting	Compared with roasting, frying leads to more loss of total lipids and UFAs in rib eye steak. In rump steak cooking, roasting retains more essential fatty acids than frying.	(Gruffat et al., 2021)
	Pork Chop	Steam cooking or griddling	Compared with pan fried pork chops, the cooked pork chops have a lower n–6: n–3 PUFA ratio.	(O'Neill et al., 2019)
	Braised pork	Slow stewing or vacuum cooking	The fat content and malondialdehyde level of stewed samples were the highest. The flavor substances of vacuum cooking samples (70 °C or 75 °C, 8 h) were similar to those of stewing samples.	(Jiang et al., 2022)
	Traditional Chinese bacon	Cold smoking for	The lipid oxidation level was the lowest	(Li et al., 2021)

Table 2 (continued)

Food category		Cooking Technology	Influence of quality	References
		3, 6, 9, 12 or 15 days	between the 9th and 12th days, and the samples smoked for the 12th and 15th days contained characteristic volatile substances such as alcohols, aldehydes, ketones and esters.	
White meat	Duck meat	Sous vide cooking (60, 70 or 80 °C and 2, 6, 12 or 24 h)	The sample boiled at 70 °C for 6 or 12 h has a higher meat and fat flavor, and has the best aroma components, such as hexanal, heptadecanal, octanal, (E) – 2-octadecenal, nonanal.	(Zhang et al., 2022)
Fish	Turbot (<i>Scophthalmus maximus</i>) muscle	Frying, baking, microwave heating, boiling in vacuum-sealed bag (BIVSB) and steaming	The loss of usable lipids and fatty acids in cooking samples was the least, and ideal flavor is produced.	(Dong et al., 2018)
	Gilthead seabream (<i>Sparus aurata</i>) and meagre (<i>Argyrosomus regius</i>) muscle	Steaming, sous-vide cooking, frying	The sensory characteristics of steam and oven cooking samples were similar. Frying led to an adverse increase in n–6/n–3 and SFA/PUFA ratio of meagre seabream.	(Alexi et al., 2019)
	European sea bass	Boiling, steaming or sous-vide cooking	The three cooking methods did not cause obvious lipid hydrolysis and oxidation.	(Nieva-Echevarria, Manzanos, Goicoechea, & Guillen, 2017)
	<i>Arapaima gigas</i> muscle	Deep fat or air frying	Air frying reduced the loss of PUFAs in the frying process, and retained better nutrition.	(Vieira et al., 2018)
	Tilapia fillets	Steaming, boiling or air frying	Steaming and boiling had similar effects on the lipids of samples.	(Li et al., 2021)
	Sturgeon steak	Frying and air frying (130 °C, 160 °C or 190 °C, 15 min)	The air fried samples have more UFAs and flavor substances (aldehydes, alcohols and esters), and lower oil content.	(Liu et al., 2022)
	Red snapper surimi	Air frying (120 °C, 160 °C or 200 °C, 10 or 15 min)	When the temperature exceeded 160 °C, the lipid was obviously	(Yu et al., 2020)

(continued on next page)

Table 2 (continued)

Food category		Cooking Technology	Influence of quality	References
Fish	Salmon muscle	Boiling, frying and grilling	oxidized and produces aldehydes and ketones. Frying and grilling increased the concentration of peroxide in the sample, and the oxidation degree of frying sample was the highest. Frying and grilling also produced n-3 PUFA oxides with biological activity.	(Leung et al., 2018)
	orange-spotted grouper (<i>Epinephelus coioides</i>)	Poaching, steaming, microwaved heating, pan-frying and deep-frying (in olive oil)	Except frying, other cooking techniques had no significant effect on the total n-3 fatty acid level of the samples. In all cooking methods, n-6 fatty acids were not lost.	(Momenzadeh et al., 2017)
	Hairtail (<i>Trichiurus lepturus</i>)	Frying (190 °C, 3, 5, 7 or 9 min) or circulating-air frying (190 °C, 12, 16, 20 or 24 min)	The lipid oxidation level of samples fried for 7 min at 190 °C and air fried for 24 min at 190 °C was low and had the best sensory quality. The characteristic flavor substances of frying were 2-heptanone, (E)-2-heptadene, 2-pentafuran and 1-pentanol. The characteristic flavor substances of air frying were 2-methylbutanol.	(Ding et al., 2022)
	shrimp (<i>Penaeus vannamei</i>)	Boiling, baking or microwaving	The samples treated with microwave had higher oxidation degree, and the contents of EPA and DHA in the samples treated with microwave and boiling were increased.	(Wang et al., 2022)
Edible oils	Extra virgin olive oil	Sauteing, deep-frying, oven baking, low temperature vacuum cooking and slow stewing	Compared with other technologies, sauteing, deep-frying and oven baking process produces more lipid oxidation products.	(Lozano-Castellon et al., 2021)
	Mashhadi melon, Iranian watermelon, pumpkin, and yellow apple seed oils	Microwave heating (1, 3, 5, 10 and 15 min)	The content of primary and secondary lipid oxidation products in various edible oils will increase with	(Hashemi et al., 2017)

Table 2 (continued)

Food category		Cooking Technology	Influence of quality	References
Edible oils	Mixed oil of palm oil and sesame oil (60:40, w/w)	Vacuum frying or frying	the extension of microwave heating time. Compared with frying, the peroxide value and free fatty acid level of vacuum frying edible oil are lower.	(Juvvi et al., 2020)
	corn oil and	Frying or low-frequency radio wave assisted frying (50 kHz)	Radio wave processing enhanced the stability of corn oil during frying.	(Hong et al., 2021)
	Soybean oil	Vacuum frying, microwave vacuum frying, ultrasonic-assisted (300 W, 600 W) microwave vacuum frying	Ultrasonic-assisted microwave vacuum frying can enhance the oxidation stability of oil in frying.	(Devi et al., 2021)
	Palm oil	Frying and infrared assisted frying	Infrared assisted frying reduced the production of free fatty acids and increased the oxidation stability of palm oil.	(Udomkun, Niruntasuk, & Innawong, 2019)

(Jiang et al., 2022). The differences of cooking temperature, time, pressure, oxygen content and other factors in different cooking technologies lead to diversified lipid oxidation degree of red meat, which further affects the quality of cooked red meat. However, more data are needed to reveal the oxidation differences and mechanisms of red meat fat under various cooking conditions.

4.2. Effects of cooking techniques on lipids in white meat and fish

Nowadays, white meat and fish have become the sources of dietary protein for more and more people. The PUFAs rich in white meat and fish not only improve its nutritional value, but also increase their sensitivity to oxidation. Therefore, the influence of cooking technology on lipid oxidation and flavor of white meat and fish cannot be ignored. Improper cooking technology selection may directly reduce their nutritional value. The effects of common cooking methods (boiling, baking and microwave heating) on lipid and volatile compounds of *Penaeus vannamei* were studied by Wang, Hu, Zhao, Liu, and Cao (2022). In their study, boiling and microwave treatment significantly reduced the content of palmitic acid, oleic acid and linoleic acid in prawns. Meanwhile, the microwave treatment group had the highest degree of lipid oxidation and significantly increased the content of volatile products of lipid oxidation such as 1-octene-3-one, pentan-1-ol, 1-propanol, octanal, heptaldehyde, valeraldehyde and hexanal in shrimp (Wang et al., 2022). Moreover, frying and baking were proven to increase the concentration of peroxides in salmon, resulting in the loss of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) (Leung, Galano, Durand, & Lee, 2018). In a vacuum low-temperature cooking research, different treatment temperature (60, 70 or 80 °C) and time (2, 6, 12 or 24 h) were arranged to investigate the sensory quality changes of duck meat (Zhang et al., 2022). The sensory results showed that cooking at

70 °C for 6 or 12 h gave duck better flavor. This may be related to the highest concentration of volatile fat oxidation products such as hexanal, heptanal, octanal, nonanal, *cis*-4-decanal, decanal, 2,4-decenal produced during cooking at 70 °C for 6 h and 12 h (Zhang et al., 2022). Moreover, Dong et al. (2018) evaluated the flavor and nutritional value of turbot muscle fried, roasted, microwave-heated, vacuum bag cooked and steamed. In their study, microwave cooking greatly increased the abundance of volatile flavor substances, such as aldehydes, alcohols and ketones, which may be produced by lipid oxidation. Nevertheless, steaming was the ideal technology to retain the UFAs in turbot muscle (Dong et al., 2018).

Recently, there have been numerous studies reporting the effects of cooking techniques on the fat composition and flavor of fish. Compared with raw fish fillet, phosphatidylcholine of steamed, boiled and fried tilapia fillet decreased significantly by 6.43 %, 8.75 % and 6.78 %, respectively (Li et al., 2021). The comparison between traditional and emerging cooking techniques is one of the hottest topics in the field. Vieira et al. (2018) compared the impacts of traditional frying and air frying on fish quality (*Arapaima gigas*). The results implied that the air frying has a fatty acid profile with higher nutritional value than traditional frying at 150 °C. At 190 °C, traditional frying resulted in 33 % increase in SFAs content and 26 % decrease in PUFAs content. Similarly, Liu et al. (2022) reported that the sturgeon fried in vacuum contains less oil but produced more lipid oxidation flavor products than the traditional fried sturgeon, such as aldehydes, alcohols and esters. A similar phenomenon was also observed by Yu et al. (2020), which may be associated with the better heat transfer effect of air frying than traditional frying. Fish is also known to be rich in *n*-3 fatty acid, previous study reported that boiling, steaming, microwave heating, pan-frying and deep-frying (in olive oil) did not change the *n*-6 fatty acid composition of grouper meat, and only deep-frying reduced the *n*-3 fatty acid content of fish meat (Momenzadeh, Khodanazary, & Ghanemi, 2017). Besides, the content of thiobarbituric acid reactive substances and peroxide value in hairtail increased with the extension of frying time, and the samples fried at 190 °C for 7 min (DF7) and air fried at 190 °C for 24 min (AF24) had the best quality (Ding et al., 2022). Aldehydes, alcohols and ketones contribute to the main flavor of the two samples. 2-heptanone, (E)-2-heptene aldehyde, 2-pentafuran and 1-pentanol are the main characteristics of DF samples, while 2-methylbutanol, ethyl methyl ketone and 3-hydroxy-2-butanone are the main characteristics of AF (Ding et al., 2022). In a word, white meat and fish might be sensitive to cooking conditions because of their rich UFAs. New cooking technologies, such as air frying and vacuum low-temperature cooking, help to retain the nutrients in white meat and fish and obtain better flavor.

4.3. Effects of different cooking techniques on edible oil

Edible oils are ubiquitous in home-cooking and are important contributors to food nutrition and flavor. The oxidation of UFA in edible oil during cooking may reduce its nutritional value and lead to health risks, such as the production of trans fatty acids and fatty acid polymers. Therefore, it is necessary to understand the roles that different cooking techniques play in the quality of edible oils to guide the selection of edible oils and cooking techniques. Lozano-Castellon et al. (2021) studied the effects of three traditional cooking techniques (sauteing, deep-frying, and oven) and three innovative cooking methods (low-temperature vacuum cooking, slow-cooker pot and vacuum pot cooking) on the composition of extra virgin olive oil. The results showed that sauteing, deep-frying, and oven produced more lipid epoxide and hydroxyl derivatives, while the innovative cooking methods caused only slight changes in olive oil composition (Lozano-Castellon et al., 2021). Besides, the trans fatty acid content of sunflower oil after stir frying and pan frying increased from 0.00 % to 1.53 % and 1.29 %, respectively. The *trans*-fatty acid content of sunflower oil after frying for 12 h was higher than that of peanut oil (Cui, Hao, Liu, & Meng, 2017). In addition,

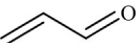
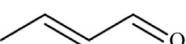
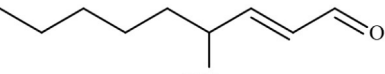
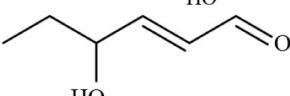
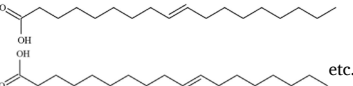
the volatile components of sunflower oil were significantly affected by frying, especially aldehydes, which were associated with lipid oxidation during frying (Multari, Marsol-Vall, Heponiemi, Suomela, & Yang, 2019). Similarly, Giuffre et al. (2020) proved the volatile components of extra virgin olive oil, pomace olive oil, palm oil and soybean oil were disturbed by cooking temperature and time. Alkaldehyde, olefinic aldehyde and dialdehyde were the most characteristic volatile components (Giuffre et al., 2020). What's more, cooking techniques may also affect the oxidation stability of oils and the changes of oxidation-related substances. The prolongation of microwave treatment time will increase the level of primary and secondary oxidation products of pumpkin seed oil, Mashhadi melon oil, Iranian watermelon seed oil and yellow apple seed oil (Hashemi et al., 2017). In another study, the PUFAs content of hemp oil dropped significantly after frying (180 °C, 60 min), while soybean oil produced the most lipid peroxides (Multari et al., 2019). Moreover, baking (180 °C for 10 min) was revealed to significantly increase the content of free fatty acids in sesame and reduce the level of PUFAs (Tenyang et al., 2017). Recently, the influence of new cooking technology on the quality of edible oil has also been studied. Compared with microwave vacuum frying (MVF), ultrasonic microwave vacuum frying (UMVF) can improve the oxidation stability of frying oil in fried mushroom slices by promoting the mass and heat transfer during frying, and reducing the generation of harmful ingredients (Devi, Zhang, & Mujumdar, 2021). Similarly, the positive effects of vacuum frying, infrared-assisted frying and microwave heating on the oxidation of palm oil, sesame oil and corn oil were proven in previous studies (Hong, Choi, Lee, & Lee, 2021; Juvvi, Selvi, & Debnath, 2020; Su et al., 2022). In general, the oxidation degree and flavor of edible oil are easily affected by cooking conditions. Choosing appropriate cooking technology according to the chemical composition of edible oil is an effective strategy to improve the nutritional value and flavor of edible oil.

5. Effects of different cooking techniques on lipid digestion

During the cooking process, food will undergo a series of physical and chemical changes. Different cooking conditions may lead to different changes in the structure of lipids, polysaccharides and proteins in foods, and may also affect the changes in the form of interaction between food components. These phenomena may contribute to changes in the sensitivity of processed foods to digestive enzymes and further regulate the bioavailability of food ingredients. It is reported that the lipid digestibility of pouched eggs was higher than that of boiled or fried eggs. This phenomenon is closely related to the least insoluble aggregates formed in pouched eggs and then the hindrance to lipid digestion was reduced (Asensio-Grau, Peinado, Heredia, & Andres, 2018). In addition, the biological accessibilities of lutein and zeaxanthin in boiled and poached eggs were higher than that in fried egg when the pancreatic enzyme secretion is insufficient (Asensio-Grau et al., 2018). In another research, homogenization and heat treatment made milk clots formed in the stomach environment have more pores, which promoted the dissolution of milk fat during digestion and improved the rate of lipid digestion (Ye, Cui, Dalgleish, & Singh, 2017). Similarly, in the process of vacuum low-temperature cooking beef tenderloin, a cooking temperature higher than 60 °C will reduce the lipid digestion rate and biological accessibility (West et al., 2021). The lipid biological accessibility after cooking at 60 °C, 70 °C and 80 °C is 15.8 %, 8.6 % and 8.3 %, respectively (West et al., 2021). This might be related to the more thorough denaturation of protein under high temperature, the formation of a more compact supramolecular structure, which hindered the contact between digestive enzymes and lipids (West et al., 2021). In another study, smoked bass was proven to have better digestion and oxidation stability than untreated samples, which might be associated with phenolic substances contained in smoked materials (Nieva-Echevarria, Goicoechea, & Guillen, 2017). Notably, different cooking methods may also lead to different levels of lipid oxidation, resulting in different levels of oxidation products such as trans fatty acids and cyclization products. Among

Table 3

Production of lipid oxidation products in cooking and health risks.

Lipid oxidation products	Chemical structure	Common sources in food	Health risks	Reference
Acrolein		Baked or fried food, frying oil	Induces the development of cardiovascular disease, liver disease, Alzheimer's disease, diabetes and tumor	(Jiang et al., 2022)
2-Butenal		Heated Rapeseed Oil, fried potato chips and doughnuts	Induces the development of tumor	(Huang & Ahn, 2019)
4-Hydroxy-trans-2-nonenal		Fried potatoes, edible oil and oil-based food rich in polyunsaturated fatty acids	Induces the development of atherosclerosis, cancer, diabetes and neurodegenerative diseases	(Liao, Zhu, & Chen, 2020)
4-Hydroxy-trans-2-hexenal		Baked or fried food, frying oil	Induces the development of cardiovascular disease	(Grootveld et al., 2020)
Malondialdehyde		Fried starch food	Induces the development of cardiovascular disease and cancer	(Karimi et al., 2022)
Trans fatty acids		High temperature baked or fried food, hydrogenated grease and butter	Induce the development of cardiovascular disease, coronary heart disease, cancer, diabetes and obesity	(Guo et al., 2022)

them, cyclized triglycerides were proven to increase the particle size of triglycerides and change the conformation of lipase. These influences hindered the enzyme activity and the contact between the enzyme and triglycerides, which lead to inhibition of lipid digestion (Li et al., 2022). However, Grau et al. (2021) reported that raw, pickled and microwave cooking did not cause differences in the lipid digestion behavior of salmon, and its lipid digestion was mainly regulated by gastrointestinal conditions. Although cooking techniques can affect the digestive behavior of lipids by causing different levels of lipid oxidation and food structure changes, future study in the area of cooking and lipid digestion needs more concern.

6. The health risks of cooking with lipids

Primary and secondary oxidation products of lipid oxidation during cooking may also pose health risks while altering the nutrition and quality of foods (Table 3). Acrolein is a thermal degradation product of glycerol, and fried food increases the exposure risk of acrolein in the diet (Jiang et al., 2022). Mona et al. (2021) observed that long-term intervention with acrolein at a dose of 5 mg/kg may damage the memory and learning ability of male SD rats by reducing the release of glutamate (excitatory neurotransmitter). Crotonaldehyde is another harmful substance produced during frying. A previous study showed that crotonaldehyde can enhance the oxidative damage of testicular cells, inhibit the activity of enzymes related to testosterone synthesis and ultimately reduce testosterone synthesis (Zhang et al., 2020). Similarly, crotonaldehyde was also revealed to induce cell apoptosis and eventually lead to lung injury through mitochondrial oxidative damage and inflammation (Li et al., 2020). In addition, the production of volatile organic compounds (VOCs) such as 2-nonenal, furan and hexanal in cooking is related to cooking conditions and food substrate. Vegetable oil rich in unsaturated fatty acids produces more VOC than lard rich in saturated fatty acids, and the presence of high temperature and water also promotes the generation of VOC (Zhang, Liu, Jia, Wang, & Han, 2019). Excessive VOC exposure increases the risk of cancer during food cooking (Zhang et al., 2019). It is noteworthy that lipid oxidation products such as malondialdehyde, epoxy fatty acid, trans fatty acid and 2-hexenal produced in cooking also have mutagenicity, genotoxicity and carcinogenicity, and may induce cardiovascular disease (Grootveld, Percival, Leenders, & Wilson, 2020). Currently, there are gaps in research on the effects of different cooking techniques on lipid health risks. However, there is already numerous evidences pointed out that repeated cooking may trigger the health-damaging effects of oils and fats. Ganesan,

Sukalingam, and Xu (2019) found that repeated heating of rapeseed oil was prone to produce a large amount of trans fatty acids and high oxidative activity products, which cause an increase in cardiovascular disease risk after ingestion. Cooking smoke from repeatedly heated edible oil contains more volatile carcinogens than fresh edible oil, such as malondialdehyde, 4-hydroxy-trans-2-nonenal, 4-hydroxyhexenal, dibenzoanthrene and benzopyrene, which are harmful to human health (Barbara, Gabriella, Gabriella, Simona, & Giuseppe, 2018; Ganesan et al., 2019; Ganesan & Xu, 2020; Vieira, Zhang, & Decker, 2017). In conclusion, the occurrence of lipid oxidation has potential impact on the health risk of lipids due to cooking. More data are needed to support the health risk assessment of lipids under different cooking methods.

7. Conclusion

Conditions such as temperature, time and oxygen pressure during cooking may lead to different degrees of oxidation of food lipid, which further affect its flavor, nutritional value, digestive characteristics and safety. Compared with traditional cooking technologies, new cooking technologies such as vacuum low-temperature cooking, ultrasonic-assisted cooking, infrared cooking and microwave cooking effectively delay lipid oxidation due to their high heat and mass transfer efficiency and low oxygen cooking environment, thereby reducing the loss of lipid quality. Therefore, the applicability of lipids in cooking foods is worth considering, which may affect the flavor, nutritional value, digestibility and safety of foods. In addition, the effects of different cooking methods on the digestive characteristics of edible oils and their mechanisms deserve further study, as well as the interactions between lipids and other food components. The combined use of various cooking technologies may be a promising means of obtaining healthy food in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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